Zengrui Han

zhan223@umd.edu
Department of Mathematics
University of Maryland, College Park
College Park, MD 20742

APPOINTMENTS

2025–2028 University of Maryland, College Park
Brin Postdoctoral Fellow

2020–2021 University of Science and Technology of China
Research Assistant

EDUCATION

Ph.D. Rutgers, the State University of New Jersey, 2020-2025
Advisor: Lev Borisov
Dissertation: GKZ hypergeometric systems and toric mirror symmetry

B.S. University of Science and Technology of China, 2016-2020

RESEARCH AREAS

Algebraic Geometry, especially mirror symmetry and related areas.

PUBLICATIONS AND PREPRINTS

2026 Minimal resolutions of toric substacks by line bundles. arXiv:2604.14384. Submitted.

2025 Hanlon-Hicks-Lazarev resolution revisited, with Lev Borisov. arXiv:2509.11077. Submitted.

GKZ hypergeometric systems and toric mirror symmetry. Ph.D. Thesis.

2024 Stringy Hodge numbers of Pfaffian double mirrors and Homological Projective Duality. arXiv:2409.17449. **Selecta Mathematica**, to appear.

Central charges in local mirror symmetry via hypergeometric duality. **Advances in Mathematics**, Volume 480, Part B (2025): 110502. arXiv:2404.16258.

2023 Analytic continuation of better-behaved GKZ systems and Fourier-Mukai transforms, **Épijournal de Géométrie Algébrique**, Volume 9 (2025), Article no. 11. arXiv:2305.12241

On hypergeometric duality conjecture, with Lev Borisov, **Advances in Mathematics**, Volume 442 (2024): 109582. arXiv:2301.01374

2019 On duality of certain GKZ hypergeometric systems, with Lev Borisov and Chengxi Wang, **Asian Journal of Mathematics**, Volume 25 (2021), No.1, 65-88. arXiv:1910.04039

TALKS AND PRESENTATIONS

2026 Geometry Seminar at George Mason University, Apr 2026.

AMS Spring Eastern Meeting 2026, Special Session on Group Actions and Combinatorics in Algebraic Geometry and Commutative Algebra, Boston College, Mar 2026.

2025 JHU-UMD Algebra and Number Theory Day, University of Maryland, Nov 2025.

Syzygies and Mirror Symmetry Virtual Seminar, Feb 2025.

2024 Rutgers Algebra Seminar, Rutgers University, Oct 2024

Columbia Enumerative Geometry Seminar, Columbia University, Sept 2024

Rutgers Algebra Seminar, Rutgers University, April 2024

2023 Rutgers Algebra Seminar, Rutgers University, Feb 2023

SERVICE AND PROFESSIONAL ACTIVITIES

Seminars

2025- Co-organizer, RIT seminar on Log Geometry, University of Maryland, College Park

2025- Co-organizer, Algebraic Geometry Seminar, University of Maryland, College Park

Refereeing

Referee for Journal of Differential Geometry.

TEACHING

University of Maryland

Math 136: Calculus for Life Sciences, Fall 2026

Math 301: Introduction to Mathematical Proof, Spring 2026

Math 402: Algebraic Structures, Fall 2025

Rutgers University

Math 551: Abstract Algebra I, TA, Fall 2024

Math 111: Precalculus, TA, Spring 2024

Math 552: Abstract Algebra II, TA, Spring 2024

Math 551: Abstract Algebra I, TA, Fall 2023

Math 477: Mathematical Theory of Probability, TA, Fall 2023

Math 244: Differential Equations for Engineering and Physics, TA, Spring 2023

Math 354: Linear Optimization, TA, Fall 2022

Math 251: Multivariable Calculus, TA, Spring 2022

Math 251H: Multivariable Calculus, TA, Fall 2021

AWARDS AND HONORS

2025 Rutgers SAS Fellowship, Rutgers University, 2025

2024 Summer PhD student Research Fellowship, Rutgers University, 2024

2019 Huang Yu Memorial Scholarship, University of Science and Technology of China, 2019

2018 Outstanding Student Scholarship, Silver Award, University of Science and Technology of China

2017 Outstanding Student Scholarship, Silver Award, University of Science and Technology of China

2015 Chinese Mathematical Olympaid (CMO), Bronze Medal, 2015

Updated October 2026